

Pleural Diseases: An Approach to a Case of Chest Pain

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: 7-Pleura_Updated.pdf (38 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; AMBOSS; Cecil 10th ed.; First Aid IM Boards 4th ed.; Osmosis

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from earlier respiratory lectures (same lecturer):** Pleural disease ties the series together. **Pneumonia** and **lung abscess** cause **empyema**; **COPD, CF and asthma** cause **secondary pneumothorax**; **PE** causes **effusion and pleurisy**; **cancer** (next lecture) causes **malignant effusion**. Pneumothorax was a complication in the Asthma lecture's Calgary flowchart.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Visceral vs parietal pleura; how fluid or air accumulates	Anatomy
Knowledge	★ Transudate vs exudate by cause	Cause lists
Knowledge	★ Pneumothorax types (spontaneous, traumatic, iatrogenic, tension)	Classification
Knowledge	★★ Light's criteria ; pleural-fluid interpretation	Numbers: 0.5 / 0.6 / $\frac{2}{3}$; pH 7.2; glucose 60/30
Knowledge	★ Management: transudate, uncomplicated vs complicated parapneumonic, empyema, pneumothorax	Drain or not; needle vs tube
Skills	Signs; CXR, US, CT, fluid labs; choose procedure by stability; ★ treat tension pneumothorax without waiting for imaging	Case vignettes
Values	Evidence-based practice; safety in conservative vs invasive choices	

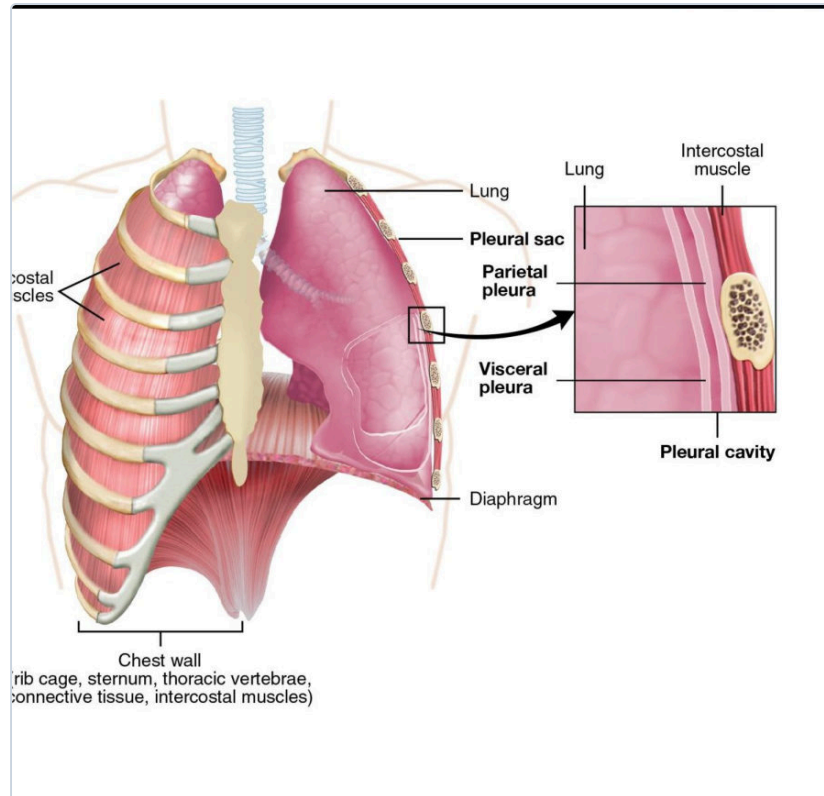
Flow: Introduction → Pleurisy → Pleural effusion (definition, types, transudate vs exudate, Light's, fluid analysis, features, diagnosis, management) → Pneumothorax (definition, classification, risk factors, features, diagnosis, size, management, procedures, algorithm) → Summary → Cases.

2. Main Content (Lecture Order)

2.1 Introduction & Pleurisy (Slides 5-7)

★ The **pleura** is a thin membrane:

- **Visceral pleura:** covers the **entire lung surface**
- **Parietal pleura:** lines the **inner rib cage, diaphragm and mediastinum**

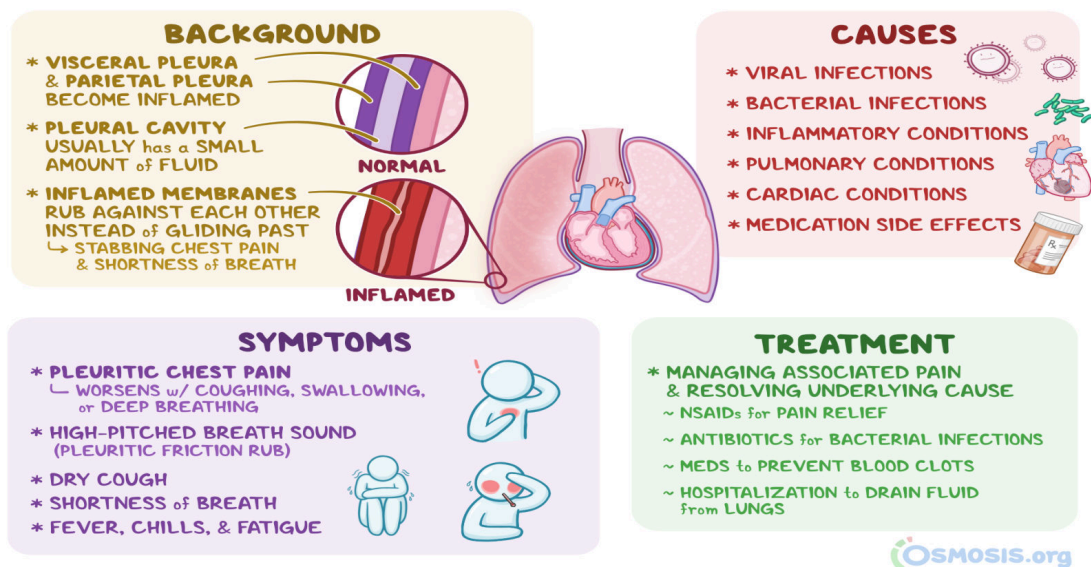


Slide 5: Visceral and parietal pleura with the pleural cavity between

★ **Pleurisy** = inflammation of the pleura.

Group	Causes
★ Viral (most common)	Adenovirus, coxsackie , CMV, EBV, influenza , parainfluenza
Bacterial	Pneumonia (parapneumonic pleuritis), TB
Inflammatory	Uraemia, SLE, rheumatoid arthritis
Pulmonary	Pneumothorax, asbestosis , malignancy (mesothelioma), ★ PE
Cardiac	MI , cardiac surgery
Drugs	Amiodarone, bleomycin, methotrexate, isoniazid, hydralazine

Features (Slide 7): ★ **pleuritic chest pain** (sharp, worse on breathing/coughing), **pleural friction rub**, dry cough, dyspnoea, fever. **Treatment:** analgesia (NSAIDs), treat the cause.



Slide 7: Pleurisy: background, causes, symptoms, treatment

🔄 **Recap: Pleurisy.** Inflamed pleura; viral is commonest; also pneumonia/TB, SLE/RA/uraemia, PE, MI, drugs. Sharp pain on breathing + friction rub.

🔗 **Link:** Inflamed or pressured pleura can fill with fluid.

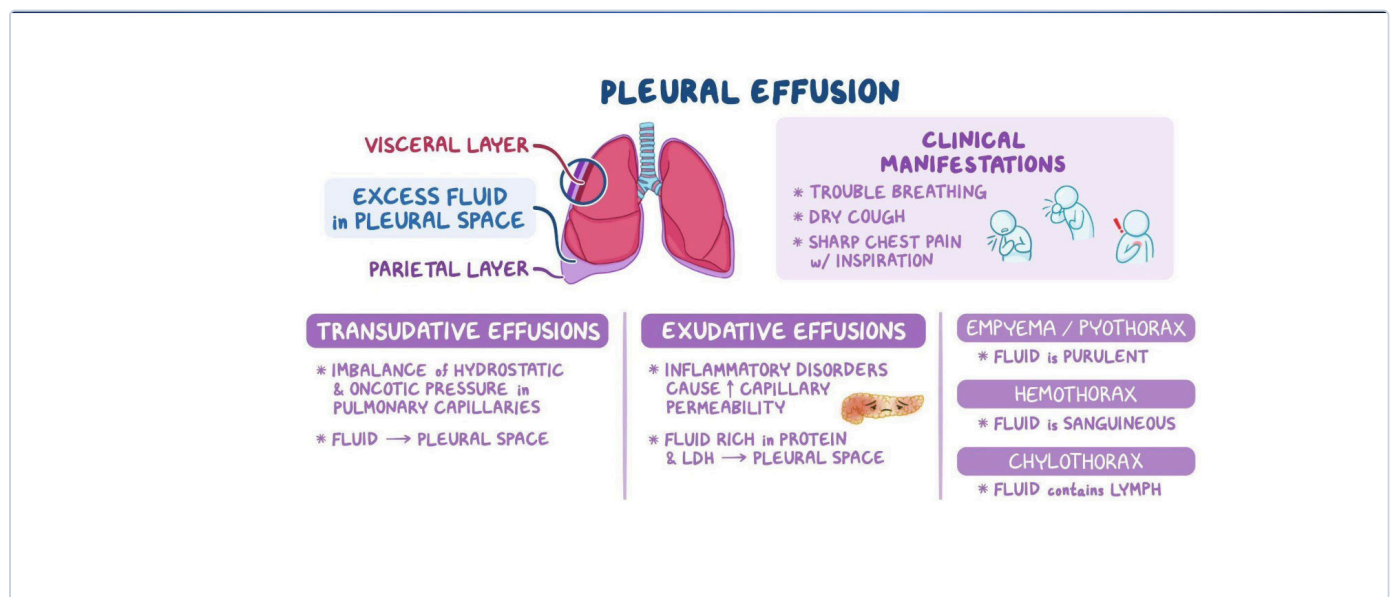
➡ **Up next: Pleural effusion.**

2.2 Pleural Effusion: Definition & Types (Slides 8-12)

★ **Definition:** excess fluid between the pleural layers that impairs lung expansion.

Mechanisms: ↑ **drainage into** the pleural space · ↑ **production** by cells in the space · ↓ **drainage out** of the space.

Fluid	Name
Blood	★ Haemothorax
Pus	★ Empyema
Chyle (lymph)	★ Chylothorax

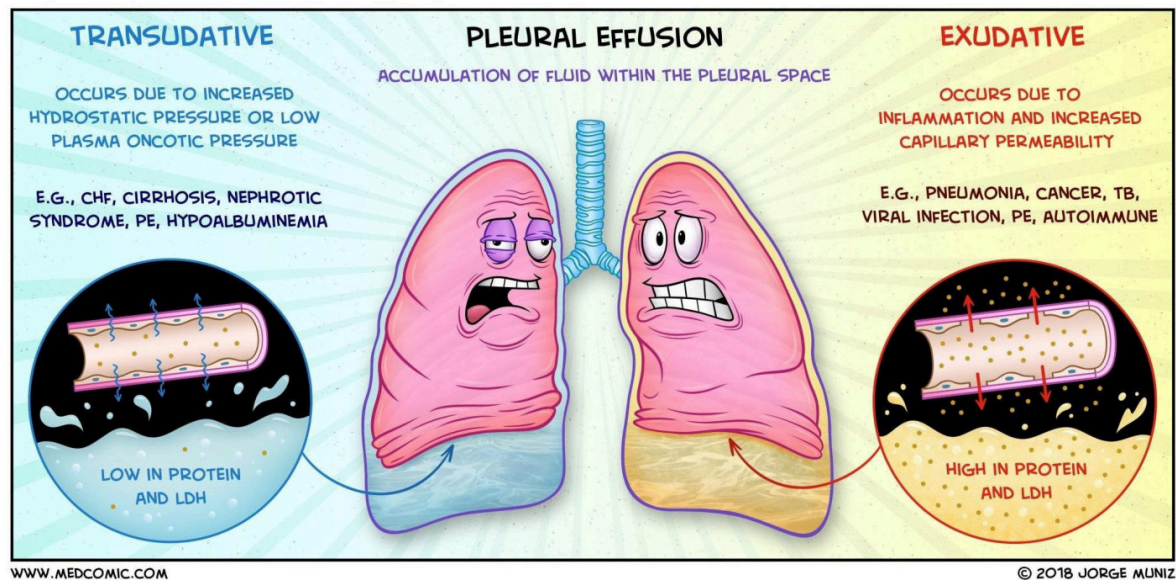


Slide 9: Pleural effusion summary: transudate, exudate, empyema, haemothorax, chylothorax

★★ Transudate vs exudate (Slides 10-11):

	Transudate	Exudate
Mechanism	↑ hydrostatic or ↓ oncotic pressure	Inflammation , ↑ capillary permeability
Protein & LDH	Low	High
★ Causes	CHF, cirrhosis, nephrotic syndrome, peritoneal dialysis, hypoalbuminaemia, atelectasis, PE	Bacterial pneumonia, TB, malignancy/metastases, viral infection, PE, collagen vascular disease
Note: PE appears in both lists.**		

MECHANISMS FOR EXUDATE:



Slide 10: Transudate vs exudate mechanisms

Other effusion causes (Slide 12):

Type	Causes
★ Empyema	Most common: pneumonia ; also infected haemothorax, ruptured lung abscess, oesophageal tear , thoracic trauma
★ Chylothorax	Trauma (incl. iatrogenic), malignancy (lymphoma, bronchogenic cancer), congenital lymphatic anomalies, thoracic duct obstruction
Haemothorax	Bleeding disorders, malignancy (mesothelioma) [added: trauma is the commonest cause]

2.3 ★★★★★ Light's Criteria & Fluid Analysis (Slides 13-14, 18)

Light's criteria (Slide 13): an effusion is an **EXUDATE** if **ANY one** is met:

Test	Transudate	★ Exudate
Protein (pleural/serum)	≤ 0.5	> 0.5
LDH (pleural/serum)	≤ 0.6	> 0.6
Pleural LDH vs serum ULN	≤ ⅔ upper limit of normal	> ⅔ upper limit of normal

Mnemonic [added]: "**0.5 - 0.6 - ⅔**": any one over = exudate.

Light's Criteria for Pleural Effusions

	Transudate	Exudate
Protein (pleural/serum)	≤ 0.5	> 0.5
LDH (pleural/serum)	≤ 0.6 Pleural LDH $\leq$ two-thirds upper limit of normal serum LDH	> 0.6 Pleural LDH $>$ two-thirds upper limit of normal serum LDH

Slide 13: Light's criteria

★★★★ Pleural-fluid interpretation (Slide 14, Table 4.10):

Test	Interpretation
★ pH	< 7.2 with parapneumonic effusion → drainage required
★ Haematocrit	$> 50\%$ of peripheral Hct → haemothorax
★ Glucose	< 60 mg/dL → complicated parapneumonic or malignancy; ★ < 30 → think RA
★ Triglycerides	> 110 mg/dL → chylothorax (milky white; thoracic duct disruption; lymphoma, cancer, trauma, lymphangioleiomyomatosis)
★ Lymphocytes	$> 50\%$ → TB or malignancy
Eosinophils	$> 10\%$ with air or blood, ★ most commonly pneumothorax; also drugs, asbestos, paragonimiasis, Churg–Strauss, PE
Amylase	↑ → acute pancreatitis, pancreaticopleural fistula, oesophageal perforation

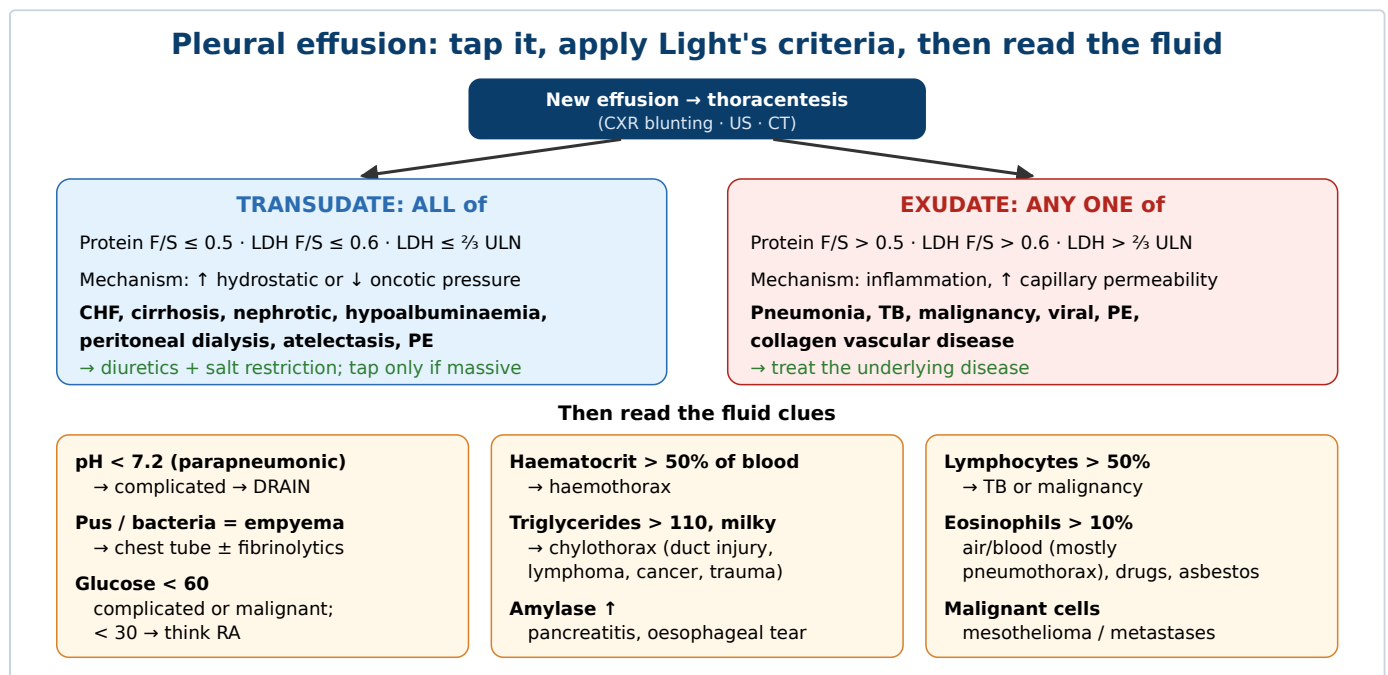
TABLE 4.10. Pleural Fluid Analysis and Interpretation

PLEURAL FLUID TEST	INTERPRETATION
pH	If pleural pH is <7.2 with parapneumonic effusion, drainage is required
Hematocrit	>50% of peripheral hematocrit suggests hemothorax
Glucose	<60 mg/dL suggests a complicated parapneumonic effusion or malignancy; if especially low (<30), think RA
Triglycerides	>110 mg/dL points to chylothorax (thoracic duct lymph disruption); milky white; due to lymphoma, cancer, trauma, or lymphangioleiomyomatosis
Lymphocytes	>50% lymphocytes is likely TB or malignancy
Eosinophils	>10% eosinophils is seen if air or blood is present, but most commonly due to pneumothorax; also consider drug reaction, asbestos, paragonimiasis, Churg-Strauss syndrome, and pleural embolus
Amylase	↑ suggests acute pancreatitis, pancreaticopleural fistula or esophageal perforation

Slide 14: Pleural fluid analysis and interpretation

★ **Work-up algorithm (Slide 18):** pleural fluid →

- **Transudate** (all criteria low) → **CHF, cirrhosis**
- **Exudate** → pH > 7.2 = **not complicated**; pH < 7.2 = **complicated** → **drainage**
- **Bacteria/pus** → **drainage**
- **Malignant cells** → **mesothelioma or metastatic cancer**



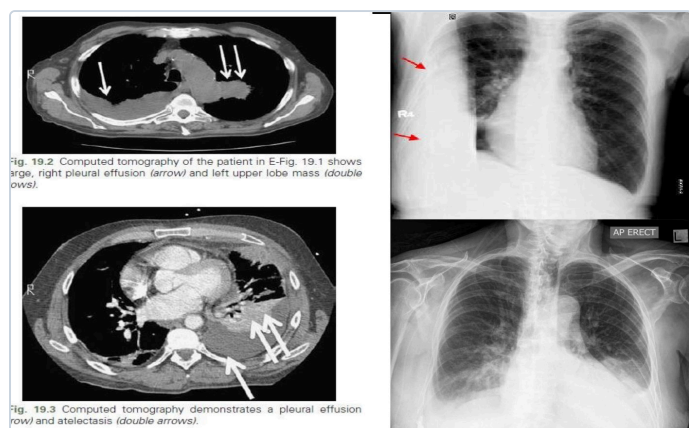
Added diagram: pleural fluid decisions

2.4 Clinical Features & Diagnosis (Slides 15-17)

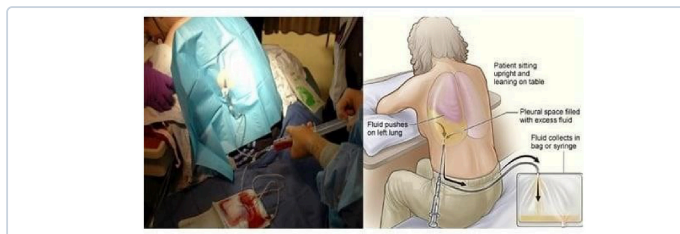
- ★ **Small effusions (< 300 mL) are often asymptomatic**
- **Dyspnoea, pleuritic chest pain, dry cough**, symptoms of the cause
- **Exam:** ★ **asymmetric expansion / lagging** on the affected side · ★ **↓ tactile fremitus** · ★ **faint or absent breath sounds** · ★ **stony dullness to percussion**

Diagnosis:

- ★ **CXR: look for blunting** (costophrenic angle) [added: about 200 mL needed on PA, about 50 mL on lateral film]
- ★ **CT and ultrasound** (US also guides the tap)
- ★ **Thoracentesis is indicated in ALL new effusions: diagnostic vs therapeutic**



Slide 16: CT and CXR of pleural effusion



Slide 17: Thoracentesis: sitting forward, needle above the rib

2.5 Management (Slide 19)

- ★ **Transudative: diuretics + sodium restriction; therapeutic thoracentesis only if massive and causing dyspnoea**
- ★ **Exudative: treat the underlying disease**
- ★ **Parapneumonic:**
 - **Uncomplicated** → antibiotics alone
 - ★ **Complicated or empyema** → chest-tube drainage ± intrapleural fibrinolytics

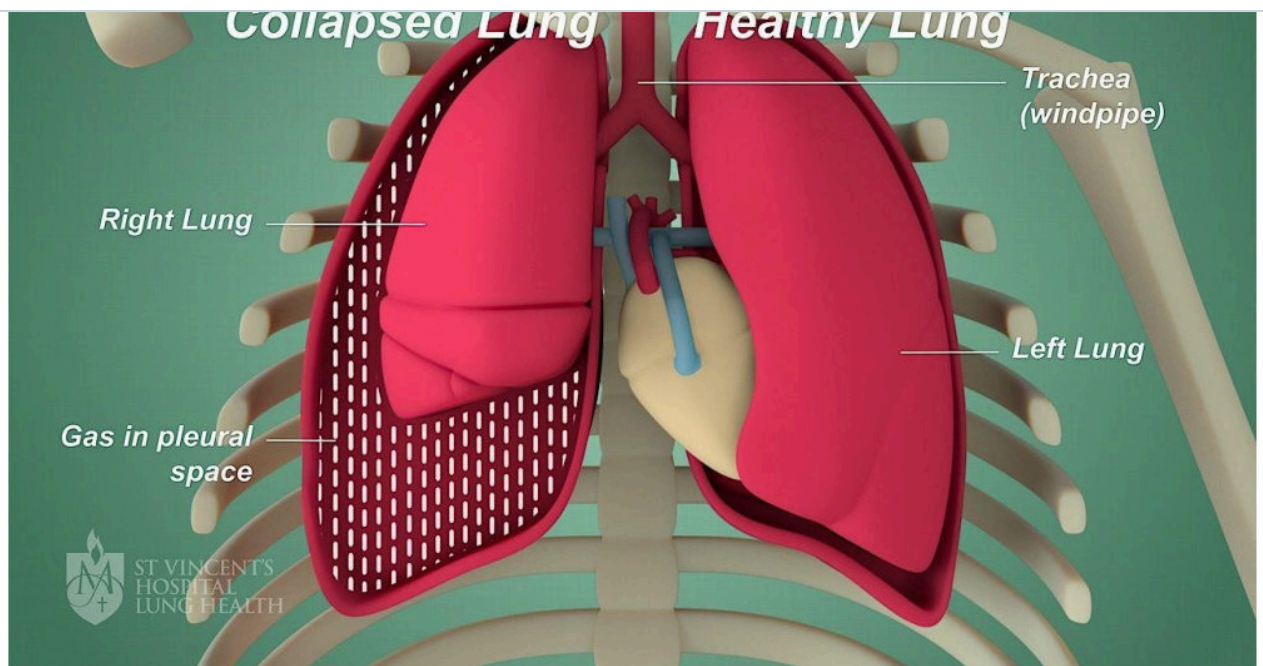
🔄 **Recap: Effusion.** Fluid in the pleural space; blood = haemothorax, pus = empyema, chyle = chylothorax. Tap every new effusion. Light's: protein > 0.5, LDH > 0.6, or LDH > $\frac{2}{3}$ ULN = exudate. pH < 7.2 or pus = drain. Glucose < 30 = RA; TG > 110 = chyle; Hct > 50% = blood; lymphocytes > 50% = TB/cancer. Transudate → diuretics; exudate → treat the cause.

🔗 **Link:** Air in the pleural space is the other big pleural emergency.

👉 **Up next: Pneumothorax.**

2.6 Pneumothorax: Definition & Classification (Slides 20-25)

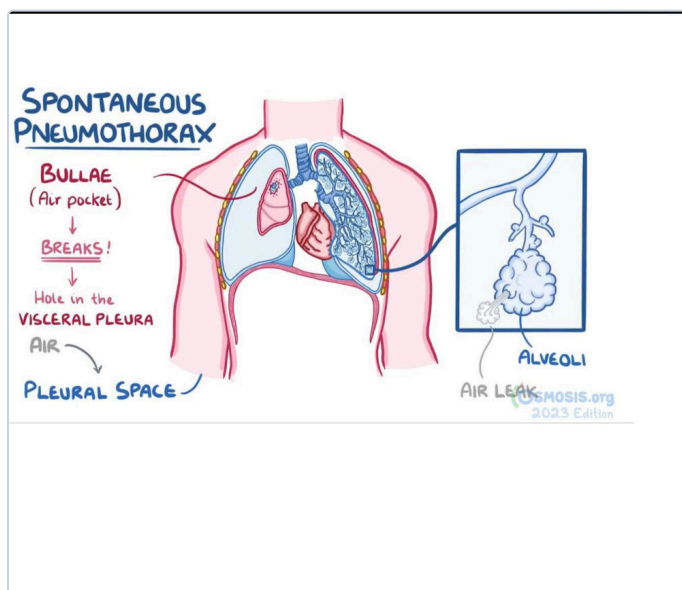
★ **Definition:** air in the pleural space; if large, the lung **collapses** and stops working.



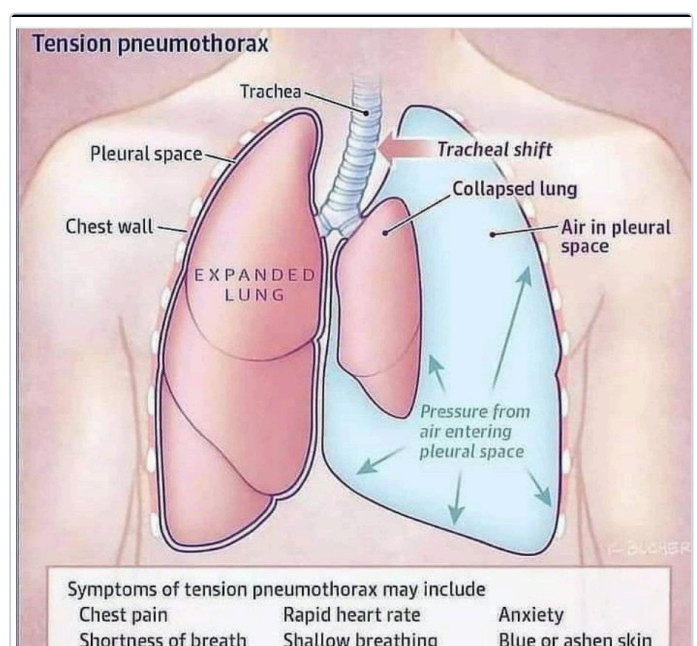
Slide 20: Collapsed lung with gas in the pleural space

★★ Classification:

Type	Features
★ Primary spontaneous	No known lung disease ; ruptured subpleural apical blebs . Risk: ★ tall, thin young men, smoking (90%; up to 20× risk, dose-related) , family history, Marfan, cocaine
★ Secondary spontaneous	Underlying lung disease : ★ COPD (ruptured emphysematous bullae), TB, Pneumocystis (cavity rupture), ★ CF , ILD , malignancy
Traumatic	Blunt (car accident) or penetrating (gunshot, stab); barotrauma from ventilation
Iatrogenic	Thoracentesis, bronchoscopy, central line , mechanical ventilation
★★ Tension	Life-threatening: progressively rising pressure in the chest → cardiorespiratory compromise



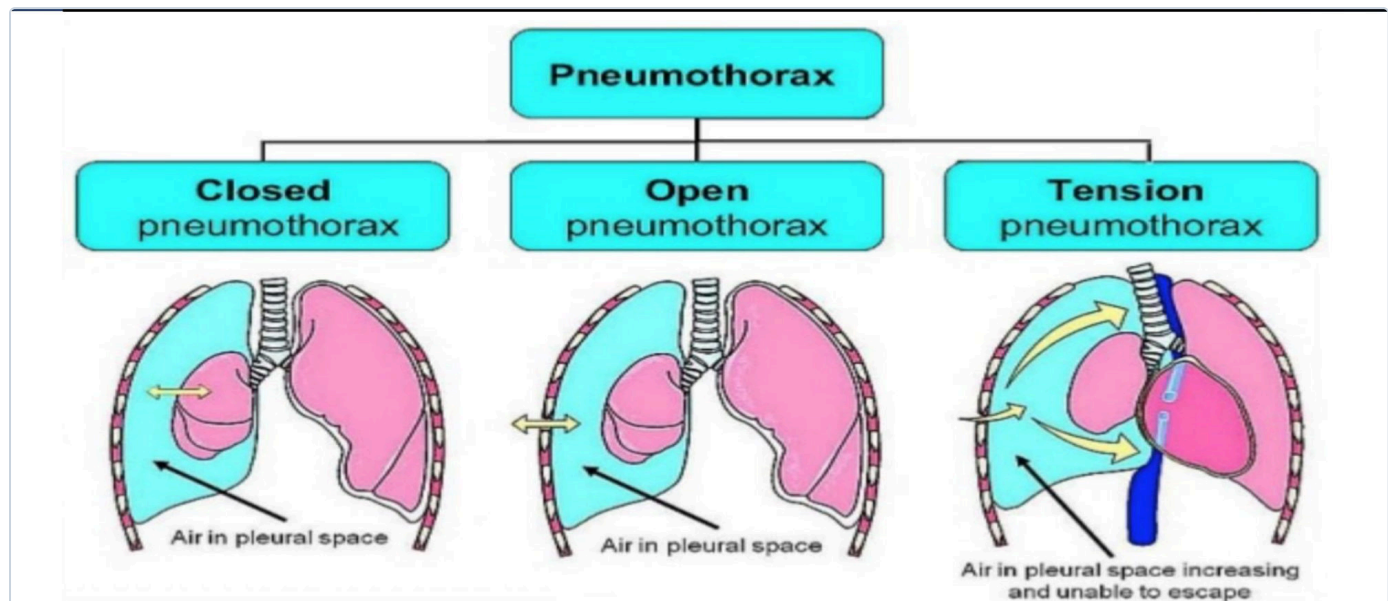
Slide 21: Spontaneous pneumothorax from a ruptured bulla



Slide 23: Tension pneumothorax: tracheal shift, collapsed lung, mediastinal compression

By mechanism (Slide 24):

Closed	Open	★ Tension
Air enters from the lung; chest wall intact	Chest-wall defect ; air moves in and out (sucking wound)	One-way valve : air enters and can't escape , pressure builds, mediastinum shifts



Slide 24: Closed vs open vs tension pneumothorax

2.7 Clinical Features (Slide 26)

- Ranges from **asymptomatic** to **haemodynamic compromise**
- ★ **Sudden, severe, stabbing, ipsilateral pleuritic pain + dyspnoea**
- ★ **↓ or absent breath sounds, HYPERRESONANCE, ↓ fremitus** on the same side

★ **Mnemonic on slide: "P-THORAX"**: **P**leuritic pain · **T**rachea deviation · **H**yperresonance · **O**nset sudden · **R**educed breath sounds (& dyspnoea) · **A**bsent fremitus · **X**-ray shows collapse

★★ Extra signs of TENSION pneumothorax:

- **Severe respiratory distress**: cyanosis, restlessness
- **Reduced ipsilateral chest expansion**
- ★ **Distended neck veins + haemodynamic instability** (obstructive shock)
- Trachea deviated **away** from the affected side *[direction added]*
- Secondary injuries (open/closed wounds)
- ★ **In ventilated patients: rapid fall in SpO₂, reduced airflow, ★ rising ventilation pressure, subcutaneous emphysema**

2.8 Diagnosis & Size (Slides 27-28)

- **Clinical diagnosis; CXR to confirm** (visceral pleural line, absent lung markings, deep sulcus sign); **POCUS**; **CT**
- ★★ **Tension pneumothorax is a CLINICAL diagnosis: don't delay treatment for imaging**
- **Size** measured on imaging; method depends on guidelines; ★ key cut-off **apex-to-cupola ≥ 3 cm = large**



Slide 27: CXR: pneumothorax with a visible pleural edge

2.9 Management (Slides 29-32)

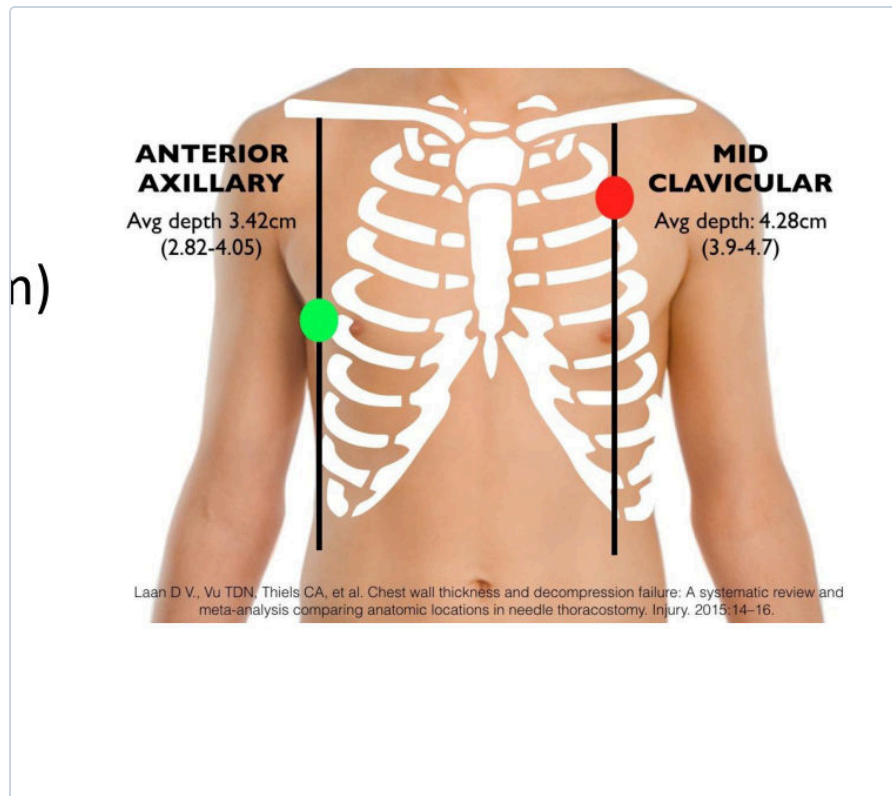
All patients: assess **stability**, give **respiratory support**, treat dyspnoea, determine **type and size**.

★★ **Unstable / high risk** (tension, bilateral, or on ventilator) → ★ **immediate chest decompression**; treat **obstructive shock**.

Situation	Management
★ Stable, small / low risk	Observation + O₂ (conservative)
Stable, larger or symptomatic	Simple aspiration or small-bore chest tube
★★ Unstable / suspected tension	★ Emergency needle thoracostomy → then chest tube immediately
Traumatic	Most need a chest tube

★★ **Procedure sites (Slide 31):**

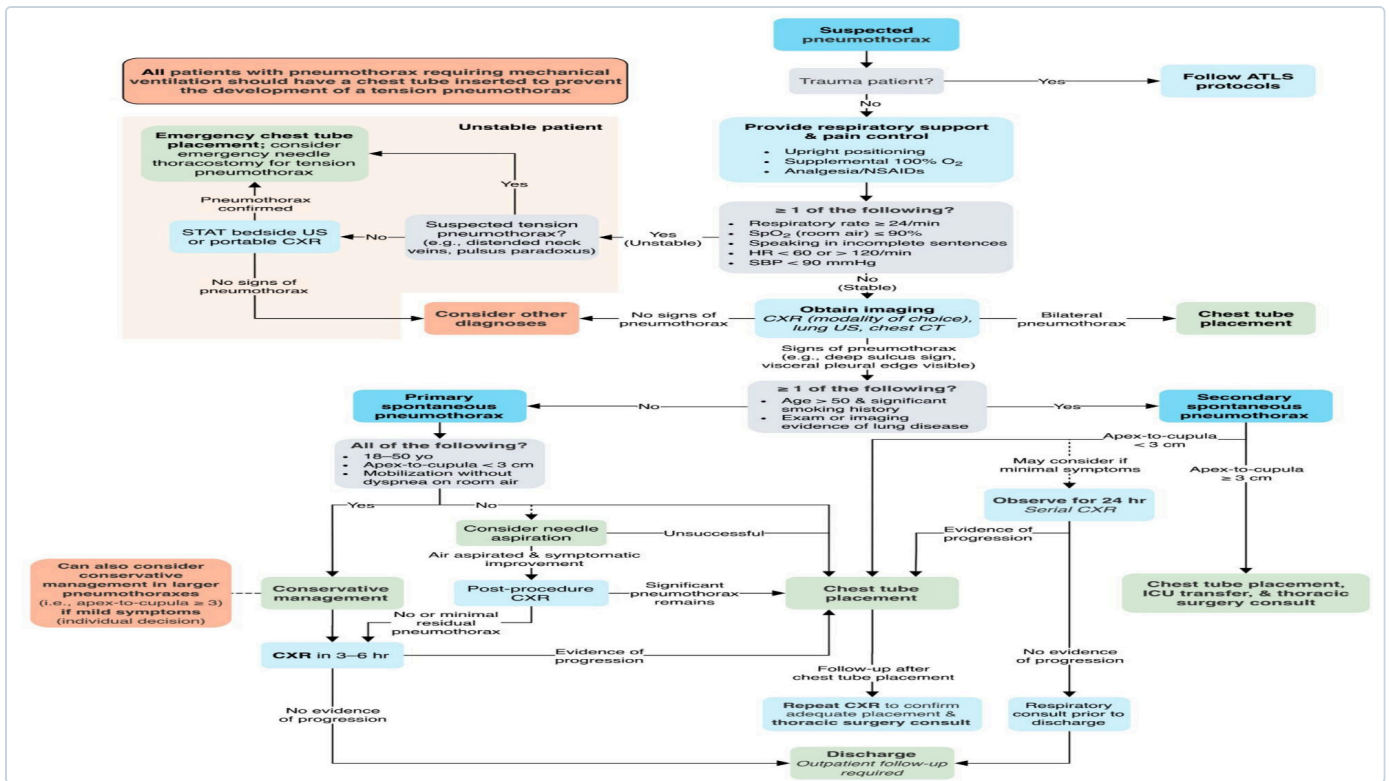
Procedure	Site
★ Needle thoracostomy	★ 2nd intercostal space, mid-clavicular line (figure also shows the 4th-5th ICS anterior axillary line as an alternative)
★ Chest tube	★ 4th-5th intercostal space (nipple line), between the anterior and mid-axillary lines (the "safe triangle" [term added])



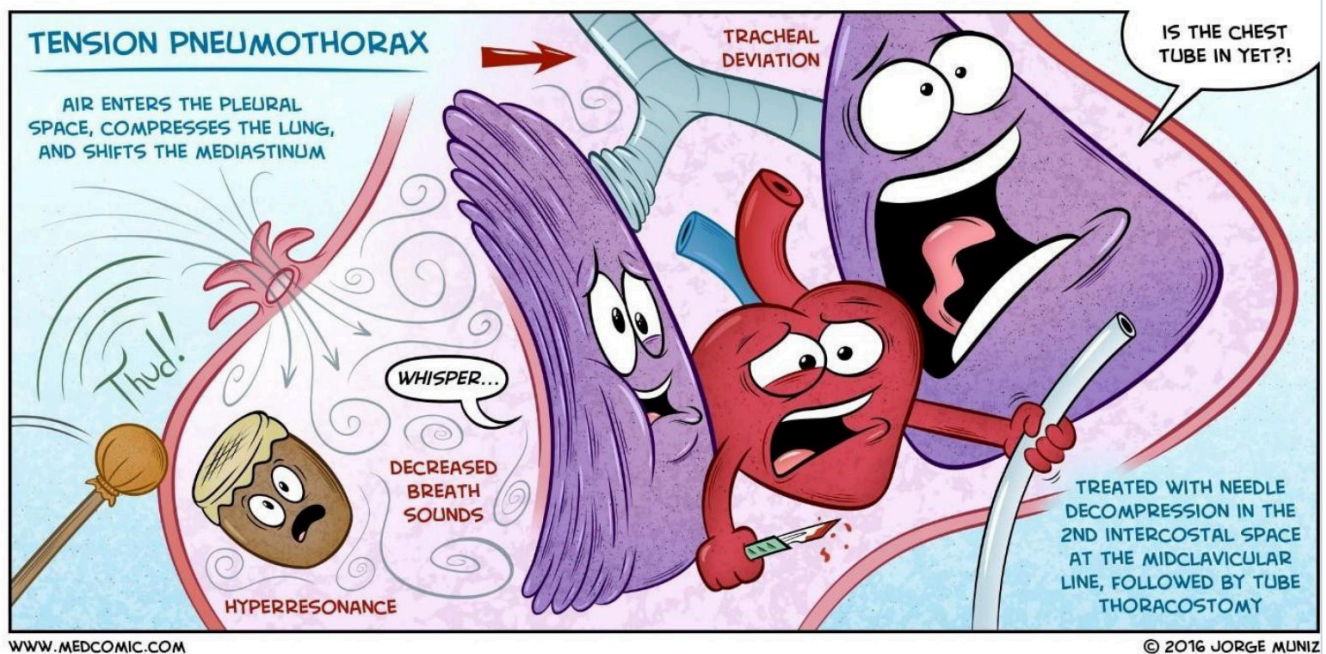
Slide 31: Needle decompression sites: 2nd ICS mid-clavicular and 4th-5th ICS anterior axillary

★★ Algorithm (Slide 32):

1. **Trauma?** → follow **ATLS**
2. Support: **upright, 100% O₂, analgesia**
3. ★ **Unstable if ≥ 1 : RR ≥ 24 , SpO₂ $\leq 90\%$, speaking in incomplete sentences, HR < 60 or > 120 , SBP < 90**
 - Suspected tension (distended neck veins, pulsus paradoxus) → ★ **emergency chest tube; consider needle thoracostomy**
4. Stable → **CXR** (modality of choice), lung US, CT; **bilateral** → **chest tube**
5. ★ **Secondary if age > 50 with significant smoking history, or lung disease on exam/imaging:**
 - **Apex-to-cupola < 3 cm** → observe 24 h with serial CXR (or chest tube); progression → tube
 - **≥ 3 cm** → ★ **chest tube, ICU, thoracic surgery consult**
6. ★ **Primary: if all of age 18-50, apex-to-cupola < 3 cm, mobilising without dyspnoea on room air** → **conservative**, repeat **CXR in 3-6 h** → discharge with follow-up
 - Otherwise → **consider needle aspiration** → if unsuccessful or significant residual → **chest tube**
7. ★ **Any pneumothorax on mechanical ventilation** → **chest tube** (prevents tension)



Slide 32: Pneumothorax management algorithm



Slide 33: Tension pneumothorax summary cartoon

Recap: Pneumothorax. Air in the pleural space. Primary (tall young smoker, apical bleb) vs secondary (COPD, CF, TB, PCP) vs traumatic/iatrogenic vs tension. Sudden pleuritic pain, hyperresonance, absent breath sounds (P-THORAX). Tension adds JVD, shock, tracheal shift, rising ventilator pressure → needle at the 2nd ICS MCL, then chest tube, without waiting for an X-ray. Stable: small → observe/O₂; large (≥ 3 cm) or secondary → aspirate or tube.

3. High-Yield Comparison Tables

Table A: Transudate vs exudate (see §2.2 and the added diagram)

Table B: Fluid clues (see §2.3)

Table C: Effusion vs pneumothorax exam

Sign	Effusion	Pneumothorax
Percussion	★ Dull (stony)	★ Hyperresonant
Fremitus	↓	↓ / absent
Breath sounds	↓ / absent	↓ / absent
Trachea (if large)	Away [added]	Away (tension)
CXR	Blunted angle, meniscus	Pleural line, no lung markings

Table D: Pneumothorax types

	Primary	Secondary	Tension
Lung	Normal (blebs)	Diseased	Any
Typical patient	Tall thin young male smoker	> 50, COPD/CF/TB	Trauma, ventilated, any
Management	Observe / aspirate	Lower threshold for tube; admit	Needle → chest tube now

Table E: Procedures (see §2.9)

4. Rules, Exceptions, Patterns

- **Tap every new effusion** (diagnostic ± therapeutic).
- **Exudate if ANY Light's criterion is met.**
- **PE can be transudate OR exudate.**
- **pH < 7.2 in a parapneumonic effusion = drain.** Pus = drain.
- **Glucose < 30 = rheumatoid**; < 60 = complicated or malignant.
- **Lymphocytic exudate = TB or cancer.**
- **Transudates are treated with diuretics, not drains** (unless massive and symptomatic).
- **Dull = fluid; hyperresonant = air.**
- **Tension pneumothorax: treat first, image later.**
- **Needle: 2nd ICS MCL. Chest tube: 4th-5th ICS, anterior-mid-axillary.**
- **Ventilated patient + pneumothorax = chest tube.**
- **Secondary pneumothorax (> 50, smoker, lung disease) is treated more aggressively.**

5. Clinical Correlations

- **Heart-failure patient with bilateral effusions** → transudate; treat with diuretics.
- **Fever + effusion after pneumonia; pH 7.0, glucose 30** → complicated parapneumonic/empyema → chest tube (Case 2).
- **Effusion with very low glucose in a woman with joint disease** → rheumatoid pleurisy.

- **Lymphocytic exudate in a migrant with weight loss** → TB (check adenosine deaminase [added]).
- **Milky fluid after thoracic surgery** → chylothorax.
- **Asbestos exposure + effusion + chest pain** → mesothelioma.
- **Tall young smoker with sudden pleuritic pain** → primary spontaneous pneumothorax.
- **COPD patient suddenly worse with unilateral hyperresonance** → secondary pneumothorax (bullae).
- **Ventilated patient: falling SpO₂, rising airway pressure** → tension pneumothorax.
- **Central line insertion followed by dyspnoea** → iatrogenic pneumothorax.

6. Rapid Revision (Last-Day)

- Visceral pleura covers the lung; parietal lines the chest wall, diaphragm, mediastinum.
- Pleurisy: **viral #1**; pneumonia, TB, SLE, RA, uraemia, PE, MI; drugs (amiodarone, bleomycin, methotrexate, INH, hydralazine).
- Effusion = excess pleural fluid; blood = haemothorax, pus = empyema, chyle = chylothorax.
- **Transudate**: CHF, cirrhosis, nephrotic, peritoneal dialysis, hypoalbuminaemia, atelectasis, PE.
- **Exudate**: pneumonia, TB, malignancy, viral, PE, collagen vascular disease.
- **Light's: protein > 0.5, LDH > 0.6, LDH > 2/3 ULN → exudate (any one).**
- **pH < 7.2 → drain · Hct > 50% → haemothorax · glucose < 60 (< 30 RA) · TG > 110 → chyle · lymphocytes > 50% → TB/cancer · eosinophils > 10% → pneumothorax/air/blood · amylase ↑ → pancreatitis/oesophageal tear.**
- Features: < 300 mL silent; dyspnoea, pleuritic pain, dry cough; ↓ expansion, ↓ fremitus, ↓ breath sounds, **dullness**.
- CXR blunting; US/CT; **tap all new effusions**.
- Rx: transudate → diuretics + salt restriction (tap only if massive); exudate → cause; uncomplicated parapneumonic → antibiotics; **complicated/empyema → chest tube ± fibrinolytics**.
- **Pneumothorax**: primary (apical blebs; tall thin male smokers, Marfan, cocaine; smoking 90%, 20x) · secondary (COPD, TB, PCP, CF, ILD, cancer) · traumatic · iatrogenic · **tension**.
- Closed / open / tension (one-way valve).
- **P-THORAX**; tension: distress, JVD, shock, tracheal shift; ventilated: ↓ SpO₂, ↑ pressure, subcutaneous emphysema.
- **Tension = clinical diagnosis → needle (2nd ICS MCL) → chest tube (4th-5th ICS, anterior-mid-axillary).**
- Unstable criteria: **RR ≥ 24, SpO₂ ≤ 90%, incomplete sentences, HR < 60/> 120, SBP < 90.**
- Small stable → observe + O₂; large (≥ 3 cm)/symptomatic → aspirate or small-bore tube; secondary ≥ 3 cm → tube + ICU + surgery; **ventilated → tube**.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 34). 23 ♂, high-speed car crash; **confused**, RR 28, **BP 91/65**, **SpO₂ 88%**; 3-cm wound at the left 5th ICS MCL; ★ **JVD**; ↓ **breath sounds and hyperresonance on the left**. Next step? **e. Needle decompression** Answer: **e.** [reasoning added; no key on the slide] Hypotension + JVD + unilateral hyperresonance/absent breath sounds = **tension pneumothorax**: a **clinical diagnosis** → **immediate needle decompression** (then chest tube). CT, CXR and echo delay treatment. Pericardiocentesis is for tamponade (JVD + hypotension but **normal** breath sounds and percussion). Thoracotomy is for penetrating trauma with arrest or massive haemothorax.

Q2 (MCQ, lecture Case 2, Slide 35). 49 ♂, alcohol misuse with recent intoxication (aspiration risk), **poor dentition**; 10 days of cough, fever 38.5 °C, right pleuritic pain; ★ **loculated right effusion** with consolidation. Pleural fluid will most likely show: **g. Glucose of 30 mg/dL** **Answer: g.**
[reasoning added] Aspiration pneumonia → **complicated parapneumonic effusion / empyema** (loculated) → **exudate with low pH (< 7.2) and low glucose (< 60)**. pH 7.5 (b), LDH ratio 0.5 (d), LDH 45 (f) and protein ratio 0.4 (h) all describe a **transudate/normal**. Amylase 200 suggests pancreatitis or oesophageal tear; Hct 30% suggests haemothorax; > 90% lymphocytes suggests TB or cancer.

Q3 (MCQ). Pleural fluid: protein ratio 0.4, LDH ratio 0.5, pleural LDH 100 (serum ULN 220). Classification and most likely cause? **Answer: Transudate** (all three criteria below the cut-offs; $\frac{2}{3} \times 220 \approx 147$) → e.g., **CHF** → diuretics.

Q4 (MCQ). Where is a chest tube inserted? a) 2nd ICS MCL **b) 4th-5th ICS between the anterior and mid-axillary lines** c) 7th ICS posterior axillary d) Subxiphoid **Answer: b.** (a) is the needle-decompression site.

Q5 (Short answer). A 22-year-old tall smoker has a 2-cm apical pneumothorax, is comfortable on room air and walking without breathlessness. Management? **Answer: Primary spontaneous pneumothorax** meeting all conservative criteria (18-50 y, < 3 cm, no dyspnoea on room air) → **conservative management** ($\pm O_2$), **repeat CXR in 3-6 h**; if stable, **discharge with follow-up** and **smoking cessation**.

8. Exam Pitfalls & Traps

1. **Exudate needs all three Light's criteria.** No, **any one** is enough.
2. **PE is always transudative.** It can be **either**.
3. **Transudates need a chest tube.** Treat the cause (diuretics); tap only if massive and symptomatic.
4. **Waiting for CXR in suspected tension pneumothorax.** Decompress first.
5. **Needle site confusion:** needle = **2nd ICS MCL**; tube = **4th-5th ICS, anterior-mid-axillary**.
6. **Dull vs hyperresonant:** fluid = **dull**; air = **hyperresonant**.
7. **Tension vs tamponade:** both cause JVD + hypotension; tension also gives **unilateral hyperresonance and absent breath sounds**.
8. **Low glucose = only empyema.** Also **malignancy** and especially **RA (< 30)**.
9. **Eosinophilic effusion = parasite.** Most commonly due to **air or blood (pneumothorax)**.
10. ⚠ **Slide inconsistency:** Slide 31 lists the needle thoracostomy indication as "stable patients with a large (≥ 3 cm)" pneumothorax and the chest tube as "unstable patients / ≥ 3 cm". Read with Slide 30: **unstable/tension → needle then tube; stable large → aspiration or small-bore tube**.
11. ⚠ The main-subtopics slide lists **lung collapse** (in brackets) but no slide covers it.